

The Machinery of Invention



Invention is not lightning; it is machinery. Five and a half centuries of patent records — and more than a century of productivity data — show recurring parts, a recognizable process, and measurable output. This paper assembles the elements, the process models, the KPIs, and the historical regularities, then extrapolates only those regularities that have survived measurement.

Invention: belief propagating across the reticulum of prior art.

LANE	METHOD
Reticulative Logic · Deep Research	Web-verified sources · red-teamed draft
EPISTEMICS	DATE
Every load-bearing claim state-tagged	2026-09-03

● TRUTH – verified against cited source

● FALSITY – documented myth / overclaim

◆ IGNORANCE – open; carries a data request

Reticulative convention: claims carry epistemic states, not vibes. Numbers appear only with a citation; where the record is silent, the paper says IGNORANCE and names the data that would settle it.

01 · DEFINITIONS

Invention, innovation, diffusion — three different events

Most
confusion
about
"measuring
invention"
starts with
using one
word for three
things. The
measurement
literature

separates
them
precisely.

Invention is
the creation
of a working
novelty — a
device,
process, or
method that
did not exist
before.

Innovation
is that
novelty
implemented:
the OECD's
Oslo Manual,
the
international
standard for
innovation
statistics,
defines an
innovation as
"a new or
improved
product or
process (or
combination
thereof) that
differs
significantly
from the
unit's
previous
products or
processes
and that has
been made
available to
potential
users... or
brought into
use," and
states the
boundary
explicitly:
*"The
requirement
for
implementation
differentiates
innovation
from other
concepts
such as
invention."* **TRUTH**

^[1] **Diffusion**
is the spread
of an

implemented innovation through a population of adopters — Everett Rogers' territory, with its S-curve and adopter categories (innovators 2.5%, early adopters 13.5%, early majority 34%, late majority 34%, laggards 16%).^[2]

The separation is old. Schumpeter built his theory of development on the "carrying out of new combinations" (1911; English ed. 1934)^[3] and in *Business Cycles* (1939) insisted that invention and innovation are distinct events — innovation can occur without new invention, and invention need not induce innovation (paraphrased; the standard citation is vol. 1, p. 84).^[4] The activity that produces inventions is itself standardized for

measurement
by the
OECD's
Frascati
Manual (first
edition 1963;
current
2015): R&D
is "creative
and
systematic
work
undertaken
in order to
increase the
stock of
knowledge...
and to devise
new
applications
of available
knowledge,"
and must be
novel,
creative,
uncertain,
systematic,
and
transferable
and/or
reproducible.
^[5]

The three events, and where measurement attaches

Synthesis of Oslo Manual 2018 · Rogers 1962 · Frascati Manual 2015 – drawn for this paper



Each transition loses candidates: most R&D does not yield patentable invention, most inventions are never implemented, most implementations never diffuse widely (\$04 quantifies the attrition). Feedback per Kline & Rosenberg.^[6]

The law's definitions came first, and they encode the same idea: Venice's 1474 statute protected "any new and ingenious device, not previously made"^[7]; England's 1624 Statute of Monopolies reserved 14-year patents to "the true and first inventor" of "new manufactures"^[8]; the US Constitution empowered Congress "to

promote the
Progress of
Science and
useful Arts"
(1787)^[9],
implemented
by the Patent
Act of 1790.
^[10] Novelty,
utility, and
disclosure
have been
the legal
skeleton of
"invention"
for five and a
half
centuries.

02 · THE
ELEMENTS

What invention is made of

Ten load-
bearing
elements,
each carried
by the
historical
record rather
than by
anecdote.
Together they
are the parts
list of the
machinery.

Strip away
the
mythology
and the
record shows
invention
behaving like
a **process**
with
recurring
parts: a
stock of prior
art,
mechanisms
that

recombine it,
institutions
that store,
steer, and
reward the
work, and
selection
environments
that decide
what
survives. The
ten elements
below recur
from the
15th-century
Venetian
arsenal to
the 21st-
century
research lab.

E1 · RECOMBINATION

New things are combinations of old things

Brian Arthur's structural analysis: "Technologies somehow must come into being as fresh combinations of what already exists" — components of technologies are themselves technologies. Schumpeter's development was already the "carrying out of new combinations." Invention grows out of the existing stock, which is why it compounds.

Arthur 2009; Schumpeter 1911/1934 ^[11]
^[3]

E2 · CUMULATION

Knowledge stacks; nobody starts from zero

Newton to Hooke, Feb 1675/6: "If I have seen further it is by standing on the sholders of Giants" — itself a 12th-century metaphor (Bernard of Chartres) that Newton was reusing. Every inventor inherits the full prior stack; the stack only grows.

Newton letter, Historical Society of Pennsylvania ^[12]

Ripe inventions arrive in multiples

Ogburn & Thomas catalogued 148 inventions made independently by two or more people; Merton argued multiples are the norm and singletons the anomaly. Bell's telephone application and Gray's caveat reached the Patent Office the same day — Feb 14, 1876. Darwin and Wallace were read jointly on Jul 1, 1858. When the prerequisites exist, someone assembles them.

PSQ 1922; Merton 1961; patent records [13][14][15][16]

Only the next step is reachable

Kauffman's term (1996/2000): at any moment, the space of buildable things is bounded by what exists now; each invention opens new doors. It explains both multiples (many people stand at the same door) and why "ahead of its time" usually means missing complements — calculus needed 17th-century mathematics, not more genius, to be found twice.

Kauffman 1996/2000; Johnson 2010; calculus priority record [17][18][19]

Markets point the search

Schmookler: capital-goods patenting tracked investment demand (railroads) — invention follows where buyers are (later re-estimation weakened, but did not erase, the pattern). Hicks: a change in relative factor prices "is itself a spur to invention... directed to economizing the use of a factor which has become relatively expensive."

Schmookler 1966 (+ critique); Hicks 1932 [20][21]

Rules that store, reward, and disclose

Venice 1474, the Statute of Monopolies 1624, the US Patent Act 1790: property in ideas, in exchange for disclosure. The Royal Society (1660) institutionalized open verification. Mokyr's "Republic of Letters" thesis — a competitive market for ideas as the origin of sustained growth — earned half of the 2025 economics Nobel, awarded "for having explained innovation-driven economic growth." Caveat from the record: Moser's world's-fair evidence shows patent laws shaping the *direction* of invention more clearly than its *level*.

Statutes; Royal Society; Mokyr 2016; Nobel 2025; Moser 2005/2013 [7][8][10][22][23][24][25]

Invention industrialized itself

Edison's Menlo Park (1876) promised "a minor invention every ten days and a big thing every six months or so." GE built the first US industrial lab (1900); Bell Labs consolidated at ~3,600 staff (1925). Individual inventors out-patented corporations until 1933; today ~86% of US-origin utility patents go to businesses. The unit of invention moved from person to system.

Rutgers Edison Papers; IEEE Milestones; Smithsonian/NSF patent-share data; Lamoreaux & Sokoloff [26][27][28][29][30]

Accidents reward the trained observer

Pasteur, Douai, Dec 7, 1854: "chance favors only the prepared mind." Fleming's contaminated plate (Sep 3, 1928), Goodyear's hot stove (1839; patent 3,633, 1844), Spencer's melted chocolate bar at a magnetron (1945), Röntgen's glowing screen (Nov 8, 1895): each accident happened in front of someone equipped to recognize it — and inside an institution able to develop it.

Wikisource 1854; ACS; USPTO; IEEE Spectrum; APS [31][32][33][34][35][36]

Technology evolves;
markets select

Basalla: continuity, diversity, novelty, selection — an evolutionary account of artifacts. Nelson & Winter: firms carry routines like genes; innovation is search; markets are the selection environment. Invention supplies variation; selection decides what becomes innovation.

Basalla 1988; Nelson & Winter 1982 [37] [38]

Sharing accelerates the
whole district

Allen's "collective invention": Cleveland-district (England) ironmasters freely revealed furnace designs, and the whole industry ratcheted forward for ~20 years. The Royal Society's journals, Mokyr's Republic of Letters, and modern open source are the same mechanism at different scales: disclosure compounds the stock everyone recombines from (E1, E2).

Allen 1983; Royal Society; Mokyr 2016 [39] [22] [23]

legal-
historical
record:
nearly
all
significan
inventions
were
made
near-
simultaneo
by
multiple
teams.
[42]

=> The
lone-
genius
account:
FALSITY –
individuals
matter,
but as
the
fastest
assemblers
of a ripe
adjacent
possible,
not as
sole
causes.

03 · THE
PROCESS

A century of process models, converging

The models of
how invention
happens were
themselves
invented,
tested, and
revised. Read
in sequence,
they converge
on one
architecture:
staged,
feedback-
ridden, and
selective.

Every serious
process
model since

1926 has kept **stages** and added **feedback**. Wallas named the cognitive stages; Usher embedded them in cumulative history; the postwar "linear model" (basic research → applied → product) turned out to be a retrospective construction that its supposed author never drew; Kline and Rosenberg replaced it with chains and loops; Cooper turned the loops into managed gates; Altshuller showed the patterns could be mined from the patent record itself.

1926

**Wallas —
the four
stages**

Preparation

→

Incubation

→

Illumination

→

Verification

(with an

"Intimation"

sub-stage,

per later re-

reading).

The

cognitive

core every

later model

keeps.^[43]

1929 / 1954

**Usher —
cumulative
synthesis**

Perception

of the

problem →

setting the

stage → act

of insight →

critical

revision.

Insight is

real but

bracketed

by

preparation

and by the

long grind

afterwards.^[44]

1943 / 1962

**Ryan &
Gross →
Rogers —
diffusion**

The Iowa

hybrid-corn

study (259

farmers,

near-total

adoption in

~13 years)

founded

diffusion

research;

Rogers

systematized

the S-curve

and

adopter

categories.^{[47][2]}

JUL 1945

**Bush —
Science,
the
Endless
Frontier**

Basic
research as
"the
pacemaker
of
technological
progress";
the report
that built
the NSF —
and got
retrofitted
into a
"linear
model" of
innovation.
[45]

1946

**Altshuller
— TRIZ**

Invention
patterns
mined from
~40,000
patent
abstracts:
40
inventive
principles,
contradiction
matrix. The
first
systematic
demonstration
that the
inventive
step itself
has
regularities.
(Larger
patent
counts
claimed by
the TRIZ
community
are
undocumented.)
[46]

1986

**Kline &
Rosenberg
— the
chain-
linked
model**

"Models that depict innovation as a smooth, well-behaved linear process badly misspecify the nature and direction of the causal factors at work." Research is called on all along the chain; use feeds back into design.
[\[6\]](#)

1986 / 1990

**Cooper —
Stage-
Gate**

The funnel operationalized: staged development with Go/Kill gates — selection built into the process rather than left to the market's end.
[\[48\]](#)

2006

Godin —
the linear
model
debunked

Historiography:
the "linear
model" was
assembled
retrospectively
in three
steps by
different
communities.
The tidy
pipeline
was never
anyone's
empirical
finding.^[49]

The
consensus
architecture

Overlaying
the surviving
models yields
one
canonical
loop — the
process this
paper
measures in
§04 and
extrapolates
in §06:

The canonical invention loop

Synthesis drawn for this paper from Wallas 1926 · Usher 1954 · Kline & Rosenberg 1986 · Cooper 1990 · Rogers 1962 – not any single author's diagram



Edison's ratio – "Genius is 1 per cent inspiration and 99 per cent perspiration" – is documented from 1901 and describes the DEVELOP & REVISE box, where the process spends most of its time.^[50]

The demand side has stages too

Diffusion is not an afterthought: it is where invention becomes economically visible, and it has measured structure. Rogers' adopter distribution:^[2]

Rogers' adopter categories — share of eventual adopters

Diffusion of Innovations (1962); percentages are Rogers' idealized normal-curve segments



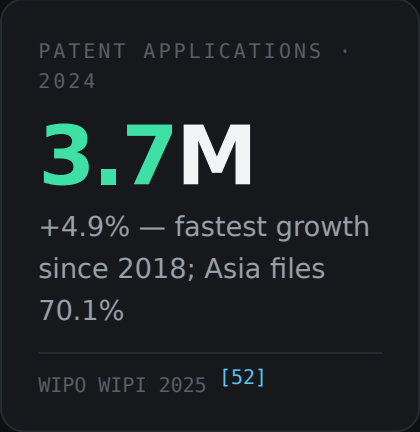
Single-hue ordinal ramp (adoption order); labels carry identity. Source: Rogers 1962.^[2]

04 · THE MEASUREMENTS

KPIs and metrics: what the dashboard actually shows

Invention is measured as a pipeline: inputs (money, people) → activity (R&D) → outputs (patents, papers, products) → outcomes (productivity). Every number below is the latest

published
value; bases
are named
because
mixing them
is the #1
measurement
error.



Basis discipline — the same world, three totals

- Global R&D 2024 is **\$3.48T** in current PPP dollars (NSF) ^[51] and **\$2.87T** in constant-2015 PPP dollars (WIPO's tracker) ^[59] — both correct, different bases. Never mix them in one series.
- China vs US: in PPP terms China's R&D (**\$1.028T**) edged past the US (**\$1.009T**) in 2024^[51]; at market exchange rates China remains roughly half the US level (OECD, March 2026 release). ^[60] Which "truth" you

get depends on the conversion you choose.

- Totals that look interchangeable rarely are: 2025 global venture funding is **\$425B** under Crunchbase's methodology and **\$512.6B** under PitchBook's — an ~\$88B gap from inclusion rules alone (both cited at the tile above). Name the methodology with the number.

Input metrics

The workhorse input KPIs are **GERD** (gross domestic expenditure on R&D), **R&D intensity** (GERD/GDP), **researchers per million**, and **risk capital**. The US spent \$937B in 2023 (est. \$993B in 2024) — split

\$138B basic research, \$174B applied, \$625B experimental development^[61] — with business funding 75% and the federal government 18%, a near-inversion of 1964 (business 31%, federal 67%).^[62] China reported ¥3.61T for 2024 (+8.3%, intensity 2.68%) and ¥3.93T for 2025 (intensity 2.8%).^[63] Government R&D budgets across the OECD fell 4.1% in real terms in 2024, with defence the main rising category.^[60]

R&D intensity — who commits the largest share of their economy

GERD as % of GDP · years differ by source and are labeled per bar



Sources: OECD MSTI Mar 2026 (Israel, Korea)^[60]; NSF (US, 2022 – latest finalized ratio)^[64]; China NBS (2024)^[63] Mixed vintages disclosed; gaps are larger than vintage drift.

Output metrics

Patents are the oldest continuous output series — the US register alone runs from Samuel Hopkins' potash patent (July 31, 1790) to ~12.7 million numbered patents by August 2026.^[53] The 2024 world totals: 3.7M applications (+4.9%), ~2.1M grants, 19.7M in force; 11.7M trademark applications; 1.22M industrial-design

applications.
[52] The PCT
route — one
international
application —
ran 275,900
filings in
2025, led by
China
(26.7%) and
the US
(19.1%). [65]

Patent
application
by
origin,
2024

Resident
+ abroad
filings
by origin
· WIPO
World
Intellectual
Property
Indicators
2025



Source:
WIPO WIPI
2025 (Nov
18, 2025).
[52] Counts
confound
inventive
output with
propensity-
to-patent –
see the
pathology
panel
below.

**Outcome
metrics —
where
invention
meets the
economy**

The outcome
layer is
productivity.
Solow's 1957
decomposition
attributed
87.5% of the
1909–1949
doubling of
US output
per hour to
technical
change [66] —
a residual
Solow
defined
broadly ("any

kind of shift
in the
production
function"),
but still the
founding
demonstration
that capital
accumulation
alone cannot
explain long-
run growth.
The modern
readings are
sobering: US
TFP growth
ran
~1.98%/yr in
1948-1972
but only
~0.87%/yr
across 1973-
2019, and
post-2004
sits below
even the
1973-96
slowdown^[67]
^[68]; Gordon
puts post-
1970 TFP
growth at
roughly one-
third of the
1920-1970
rate.^[69]

**A natural
experiment
in metric
behavior:
pharma**

Drug R&D is
the best-
instrumented
invention
pipeline on
earth, and it
shows both
metric
pathologies
and real
signal.

**Eroom's
Law**

(Moore's,
backwards):
approved
drugs per
inflation-

adjusted
billion dollars
of R&D
halved
roughly every
9 years from
1950 to 2010
— an ~80-
fold
efficiency
decline^[70] —
before
breaking
upward
around 2010.
^[71] Cost per
approved
drug:
\$2.558B pre-
approval
(Tufts, 2013
dollars)^[72];
Deloitte's
cohort
average ran
\$2.23B with
5.9% IRR in
2024,
improving to
\$2.67B with
7.0% IRR in
2025.^[73]

**FDA
novel
drug
approvals
(CDER),
2021-
2025**

Novel
approvals
per
calendar
year –
the
pipeline's
output
counter



Source: FDA
CDER novel
approvals;
2025
summary via
RAPS.^[74]
Output
holds ~37–
55/yr while
cost-per-
asset and
returns
move – a
reminder
that count
metrics and
efficiency
metrics
answer
different
questions.

Firm-level
KPIs —
the
practitioner's
panel

Corporate
R&D
concentrates
hard: the
world's top
2,000
corporate
investors
spent

€1,446B in
2024 — over
90% of all
business-
funded R&D
— with US
firms
carrying
47.1% of it;
Amazon
entered the
ranking at #1
(~€65B, a
Scoreboard
estimate),
and FY2024
filings show
Alphabet at
\$49.3B and
Meta at
\$43.9B.^{[75][76]}
The standard
firm KPI
panel:

- **R&D intensity**
(R&D / net
sales):
pharma &
biotech
lead at
~16%,
software &
computer
services
~12% (EU
Scoreboard
sector
data;
edition
vintages
vary). IGNORANCE
— current-
edition
sector split

not
verified;
treat the
levels as
indicative.
[\[75\]](#)

- **New Product Vitality Index** (3M, 1988): share of revenue from products launched in the past five years; 3M's benchmark ~30%.[\[77\]](#)

- **RQ (Research Quotient):** Knott's estimated output elasticity of R&D — the closest thing to a firm-level "return on invention."
[\[78\]](#)

- **Funnel metrics:** stage-gate pass rates, time-to-market (no standardized definition — start and end points vary by firm), patents per R&D dollar.[\[48\]](#)

- **Composite indices:** the Global Innovation Index 2025 (139 economies) ranks Switzerland #1 for the

15th
consecutive
year;
China
entered
the top 10
for the first
time. GII
2026 is
due Sep
29, 2026
— after
this paper.
^[79]

only
~5% of
US
patents
are
ever
litigated
or
licensed
for a
royalty
(~1.5%
litigated)
and
"perhaps
95%"
are
never
valuable
enough
for
either
(Lemley,
ca.
2001).
[\[82\]](#)

[IGNORANCE]
The
current
patent→val
conversion
rate.
DATA
REQUEST:
a
post-
2001
replicati
of
Lemley's
licensing
rates
on
modern
USPTO
cohorts.

=>
Invention
is
measurable
– **as a
battery
of
imperfect
gauges**
(inputs +
outputs +
outcomes),
never as
any
single
number.

Thirteen things history

actually established

Each lesson is
stated as a
claim, with its
evidentiary
basis and
epistemic
state. These
thirteen are
the premises
the trajectory
section is
allowed to
extrapolate
from.

01

**Invention
is
recombination,
so
it
compounds**

TRUTH
New
technologies
are
combinations
of
existing
ones;
each
addition
enlarges
the
space
of
possible
next
combinations.

This
is
why
the
process
accelerates
across
centuries
rather
than
exhausting
itself.

Arthur
2009;
Schumpeter
1911/1934;
Kauffman's
adjacent
possible
[31]

Ripe
inventions
arrive
on
schedule,
in
multiples

TRUTH

148

documented
independent
co-
inventions
by
1922;
same-
day
telephone
filings
(Feb
14,
1876);
Darwin-
Wallace
read
jointly
(1858).
When
prerequisites
and
demand
are
present,
the
invention
is
close
to
inevitable
—
the
interesting
variable
is
who
assembles
it
first,
not
whether.

Ogburn
&
Thomas
1922;
Merton
1961;
Lemley
2012
[4a]

The lone genius is bad history

FALSITY of the myth

The canonical solo inventors dissolve on inspection ("nearly a dozen" incandescent-lamp inventors in the 1870s), and knowledge work has shifted decisively to teams: mean S&E team size grew from under 2 to ~3.5 authors in 45 years, and teams produce the most-cited work.

Lemley 2012; US DOE lamp history; Wuchty, Jones & Uzzi 2007 [40]

Institutions

are

the

operating

system

TRUTH

Patents

(1474

→

1624

→

1790)

bought

disclosure

with

temporary

exclusivity;

open-

verification

societies

(1660)

made

knowledge

cumulative

and

trustworthy;

the

2025

economics

Nobel

was

awarded

precisely

"for

having

explained

innovation-

driven

economic

growth"

through

institutions

and

creative

destruction.

Nuance:

patent

laws

demonstrably

steer

invention's

direction;

their

effect

on

its

level

is

contested.

Venice

1474;

Statute

of

Monopolies;

Royal

Society;

Nobel

2025;

Moser

2005/2013

[29]

Invention
industrialized:
person
→
lab
→
team
→

system

TRUTH

Menlo
Park
scheduled
invention
("a
minor
invention
every
ten
days");
GE
(1900)
and
Bell
Labs
(1925)
made
it
a
corporate
function;
individual
inventors
out-
patented
firms
until
1933,
while
~86%
of
US-
origin
utility
patents
now
go
to
businesses.
The
production
function
of
invention
is
an
organizational
variable,
not
a
constant.

Rutgers
Edison
Papers;
IEEE
Milestones;
Smithsonian/NSF
shares;
Lamoreaux
&
Sokoloff
[20]

Demand
and
relative
prices
steer
the
search

TRUTH – direction; magnitude debated

Railroad
patenting
tracked
railroad
investment
(Schmookler);
dear
factors
invite
economizing
inventions
(Hicks
1932).
Re-
estimation
weakened
Schmookler's
correlations,
so
the
lesson
is
directional
steering,
not
demand-
determinism.

Schmookler
1966
+
Research
Policy
critique;
Hicks
1932
[20]

Impact
arrives
decades
after
invention

TRUTH

Electric
motors
were
<5%
of
US
factory
drive
power
in
1899
—
17
years
after
Edison's
first
central
station
—
and
passed
50%
only
in
the
early
1920s,
when
the
productivity
surge
finally
appeared.
Solow's
1987
quip
("the
computer
age
everywhere
but
in
the
productivity
statistics")
marked
the
same
lag
for
IT;
the
J-
curve
formalizes
it
via
intangible
complements.

David
1990;
Solow
1987;
Brynjolfsson,
Rock
&
Syverson

Science
and
invention
have
fused

TRUTH

Patents

citing

science

rose

from

6.7%

(1976)

to

25.6%

(2018);

61%

of

US

patents

trace

back

to

a

research

article

and

80%

of

cited

articles

link

forward

to

a

patent.

Meanwhile

large

firms

retreated

from

doing

science

themselves

(publishing

firms

17%→6%,

1980-

2006)

—

buying

the

goose's

eggs

while

releasing

the

goose.

Marx

&

Fuegi

2020;

Ahmadpoor

&

Jones

2017;

Arora,

Belenzon

&

Patacconi

Each
idea
costs
more
than
the
last

TRUTH – with a live dissent

Aggregate
US
research
productivity
fell
×41
since
the
1930s
(≈−5.1%/yr);
a
Moore's-
law
doubling
now
takes
>18×
the
researchers
of
the
early
1970s;
age
at
first
invention
rose
~0.6
years/decade
as
the
knowledge
burden
grew;
drug
R&D
efficiency
halved
every
~9
years
for
six
decades
before
breaking
upward
around
2010.
Dissent
exists
(Census
WP
25-
29:
patents
per
R&D
input
rising)
and
is
flagged

Growth
held
because
effort
scaled

TRUTH

The response to falling research productivity was more researchers and more money: global R&D reached \$3.48T (2024), and in PPP terms China's spend passed the US in 2024 — the first change at the top since the series began. Scaling inputs is the historical answer to declining idea-yield.

Bloom et al. 2020 (effort doubling); NSF 2026 [89]

Cost curves are the most predictable regularity we have

TRUTH

Wright's 1936 aircraft learning curve generalizes: across 50+ technologies, cost declines follow statistically forecastable exponentials. Solar modules fell \$106/W (1976) → ~\$0.38/W (~99.6%), at a 20.2% learning rate per doubling of cumulative capacity. Where a deployment-driven curve exists, forecasting invention's *effects* becomes tractable even when individual inventions stay unpredictable.

Wright 1936; Farmer & Lafond 2016; Way et al. 2022; OWID [98]

Serendipity
is
a
harvest,
not
a
lottery

TRUTH
Penicillin,
vulcanization,
X-
rays,
the
microwave:
each
"accident"
landed
on
a
trained
observer
inside
an
institution
able
to
develop
it.
Pasteur
said
it
in
1854;
the
20th-
century
lab
systematized
it.
Slack
time
and
exploratory
capacity
are
inputs,
not
waste.

Pasteur
1854;
Fleming
1928;
Röntgen
1895;
Spencer
1945;
3M
Post-
it
[30]

13
**Openness
accelerates
the
whole
district**

TRUTH

Cleveland's
ironmasters
ratcheted
an
entire
industry
by
publishing
furnace
designs;
the
Republic
of
Letters
did
it
for
Enlightenment
Europe;
patents
themselves
are
a
disclosure
bargain.
Sharing
enlarges
the
common
stock
that
everyone
recombines
from
—
the
compounding
term
in
Lesson
1.

Allen
1983;
Mokyr
2016;
Royal
Society
[29]

Estimating forward without inventing

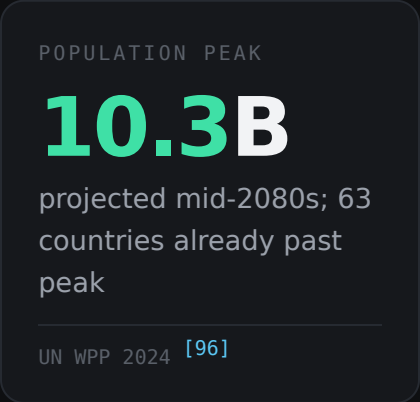
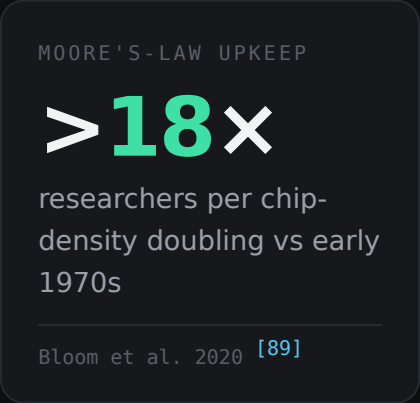
the future

This section extrapolates only the regularities established in §05, reports published estimate ranges where experts disagree, and names the data that would falsify each projection. It is an extrapolation, not a prophecy.

6.1 · The scissors: more effort required, fewer hands available

Two measured curves are closing on each other. The first is research productivity: holding growth constant has required doubling research effort every ~13 years.^[89] The second is demography: the researcher pipeline ultimately draws on population, and the UN

projects
world
population to
peak around
10.3 billion in
the mid-
2080s — it
has already
peaked in 63
countries.^[96]
Chad Jones'
growth
theory makes
the collision
explicit: with
sub-
replacement
fertility, idea-
based growth
models
converge to
an "Empty
Planet"
outcome of
stagnating
living
standards.^[97]



(derived here, shown in full): the 13-year doubling requirement implies effort growth of $2^{1/13} - 1 \approx$

5.5%/yr. UNESCO's series shows researchers per million rising 1,141 (2015) →

1,486 (2023)^[58]: a ratio of 1.302 over 8 years, i.e.

$1.302^{1/8} - 1 \approx$ **3.4%/yr.**

The per-capita series is thus growing ~2 points/yr slower than the requirement.

IGNORANCE The precise absolute-headcount gap.

DATA REQUEST: UNESCO UIS absolute FTE researcher series 2015–2023 (latest verified point: 8.854M FTE in 2018^[57]) to compute total-effort CAGR against the 5.5%/yr requirement.

A live dissent must be carried, not buried: a 2025 Census Bureau working paper finds patents per R&D input *rising* — "Ideas Are Not Getting Harder to Find."^[91] The disagreement is about how

to measure
idea output;
§06's
projections
therefore
lean on the
scissors only
for direction,
never for a
point value.

**6.2 · The
counterforce:
invention
of new
methods
of
invention**

The historical
precedent is
exact.
Whitehead,
1925: "The
greatest
invention of
the
nineteenth
century was
the invention
of the
method of
invention."^[98]
Griliches
showed
hybrid corn
was "the
invention of a
method of
inventing"
(1957)^[99];
Cockburn,
Henderson &
Stern argued
in 2018 that
deep learning
is the next
general-
purpose
method of
invention.^[100]
Since then
the evidence
has moved
from
argument to
record:

NOV 2020

**AlphaFold
2 solves
the
CASP14
benchmark**

Recognized
by CASP
organizers
as a
solution to
the 50-year
protein-
structure
problem;
Nature
paper Jul
2021.^[101]
^[102]

JUL 2022

**The
protein
universe,
predicted**

AlphaFold
DB expands
to
214,684,311
structures
— "nearly
all
catalogued
proteins
known to
science."
^[103]

NOV 2023

Materials
at scale
— with a
caution

GNoME
predicts
2.2M
crystals
(~380k
stable; 736
realized
concurrently).
[104] The
2024
*Chemistry
of Materials*
critique
found
"scant
evidence"
of
compounds
meeting
"the trifecta
of novelty,
credibility,
and utility"
— scale is
not yet
usefulness.
[105]
GraphCast
beats the
leading
physics
weather
model on
~90% of
1,380
targets the
same
month. [106]

OCT 2024

The
Nobel
threshold

Chemistry
Nobel: half
to Baker
"for
computational
protein
design,"
half jointly
to Hassabis
and Jumper
"for protein
structure
prediction."
AI-assisted
invention
enters
science's
highest
canon. [107]

FEB–MAY
2025

Hypothesis machines

Google's multi-agent "co-scientist" reproduces an unpublished, experimentally confirmed superbug mechanism among its top-ranked hypotheses.^[108] AlphaEvolve improves on Strassen's 1969 matrix-multiplication record (48 multiplications for 4×4 complex), recovers ~0.7% of Google's fleet compute, and improves best-known solutions on ~20% of ~50 open problems.^[109] FutureHouse's Robin proposes a drug-repurposing candidate with humans running the wet lab.^[110]

JUN 2025

First
generative-
AI-
discovered
drug
clears
Phase IIa

Rentosertib
(generative-
AI target +
molecule)
publishes
randomized
Phase IIa
results in
*Nature
Medicine*:
+98.4 mL
FVC (60
mg) vs
−20.3 mL
placebo in
IPF; Phase
III began Jul
2026.^[111]
^[112] As of
mid-2026,
**no end-to-
end AI-
discovered
drug is
FDA-
approved.**^[113]

JUL 2025

IMO gold,
certified

Gemini
Deep Think
scores
35/42 —
gold-medal
standard,
officially
graded by
IMO
coordinators
(OpenAI
claimed the
same score,
graded
privately,
uncertified).
One year
earlier AI
stood at
silver; two
years
earlier, at
geometry-
only.^[114]
^[115]^[116]

MAR 2026

The referee joins the loop

Sakana's AI-Scientist line — whose workshop paper passed peer review in 2025 under organizer supervision — publishes in *Nature*, including an automated reviewer that matches human judgments. Generation *and* verification are both automating. ^[117]

JUL 2026

Reported, not yet established

IGNORANCE
Press reports say multiple AI systems scored perfect 42/42 at IMO 2026 (Shanghai). Secondary reporting only; not independently confirmed for this paper. DATA REQUEST: official IMO 2026 grading records. ^[118]

Note what the timeline does *not* show: an approved AI-discovered drug, an AI-designed

material in
mass
production,
or an
autonomous
system
running the
full loop of
§03 —
including the
wet lab and
the market
gate —
without
humans. The
pattern so far
matches
§05's Lesson
7: the
SEARCH and
INSIGHT
boxes are
automating
fastest;
DEVELOP-
and-REVISE
and SELECT
still run at
human and
regulatory
speed. That
is exactly
where
electrification
stood in 1899
— capability
demonstrated,
complements
pending. TRUTH as description; the electrification analogy is an interpretive frame, not evidence

**6.3 · What
the
credible
estimates
disagree
about**

Institutions
that have
modeled AI's
aggregate
effect
disagree by
roughly an
order of
magnitude.
Rather than
pick a
winner, the
paper plots

the published
range,
normalized to
an
approximate
annualized
boost
(derivations
in the table):

Published estimates of AI's boost to annual growth

6.4 · The input curves the the projections ride on

FRONTIER TRAINING
COMPUTE

4-5x/yr

growth 2010-2024
≈4.1x/yr; ~4.4x/yr
since 2022

Epoch AI [124]

AI TASK-LENGTH
DOUBLING

~7 mo

human-time length of
tasks agents complete
at 50% success, ~6-yr
trend

METR 2025 [125]

DATA-CENTER
ELECTRICITY

945
TWh

IEA 2030 projection,
from ~415 TWh (2024)

IEA, Energy & AI 2025 [126]

EXPERT HLMI MEDIAN

2047

50% probability year,
per 2,778 AI authors
(2023 survey; 10% by
2027)

Grace et al. 2024 — a survey
aggregate, not a fact about
the future [127]

Two further steering currents, both measured: **geography** — China passed the US in PPP R&D (2024) [51], leads the Nature Index (Share 32,122 vs 22,083, 2024 data) [128], and entered the GII top 10 (2025) [79], so the next method-of-invention race is genuinely multipolar; and

energy/materials

cost curves

— Wright's-

law dynamics

(solar:

−99.6%

since 1976 at

a 20.2%

learning

rate^[95])

remain the

best-

validated

forecasting

instrument

available^[93]

^[94], and now

apply to the

compute

build-out

itself.

Forecast
humility
— the
base
rate of
confident
prediction

- FALSITY

Watson's 1943 "world market for maybe five computers" — apocryphal; no primary evidence exists (it grew from a misread 1953 remark about one machine).
[129]

The most-quoted forecasting failure in tech history never happened.

- Ken Olsen's 1977 "no reason for any individual to have a computer in his home" was real but concerned home-automation mainframes, and he protested the decontextualization.
[130]

- TRUTH

Wilbur

Wright, 1908, on himself: "In 1901 I said to my brother Orville that man would not fly for fifty years. Two years later we ourselves made flights."^[131]

Even the inventors misforecast their own field — in the pessimistic direction.

Hence this section's method: extrapolate measured trends, carry published ranges, name falsifiers — and treat every named-year forecast, including the survey medians above, as a claim about forecasters, not about the future.

6.5 · The
cast
verdict

of
Materials
critique.

[IGNORANCE]

Timing
of
research-
capable
machine
intelligen

DATA
REQUEST:
none
can
settle
it
now –
the
2047
HLMI
median
is an
aggregate
of
2,778
forecasts
and
the
13-
year
inward
jump
between
the
2022
and
2023
surveys
shows
how
unstable
such
medians
are.
Treat
as
sentiment
data.

CONTRADICTION

Bloom
et al.
(research
productivi
falling
×41) ↓
Census
WP 25-
29
(returns
to R&D
rising)
–
unresolved
measuremen
dispute;
severity
moderate.
Both
retained.
[\[89\]](#)[\[91\]](#)

=>

Direction:
TRUTH-
grade
extrapolation

Invention
continues
to
industrialize
into
automated,
compute-

fed
search
while its
human-
headcount
input
flattens;
the
binding
constraints
shift
from
researchers
toward
compute,
energy,
and –
hardest
of all –
verification
and
selection
capacity
(the
SELECT
box of
\$03,
which
regulators,
wet labs,
and
markets
still run
at human
speed).
Magnitude:
IGNORANCE,
bounded
by the
published
0.1–1.5
pp/yr
range.
Every
premise
above is
falsifiable
by the
named
data,
most of
it
arriving
before
2031.

How this paper knows what

it

claims

The
apparatus,
disclosed.
What was
verified, how,
and where
the edges of
confidence
sit.

Method

This paper
was
researched
on **2026-09-03** by three
parallel deep-
research
passes — (a)
the history
and theory of
invention, (b)
measurement
systems and
current KPI
values, (c)
evidence
bearing on
the future —
each
instructed to
verify every
claim against
primary or
authoritative
sources
(WIPO, OECD,
NSF NCSES,
UN DESA,
journal and
publisher
pages, the
Nobel
Foundation,
Our World in
Data) and to
mark
anything
unverifiable
as unusable.
The
assembled
draft was
then
adversarially

red-teamed
claim-by-
claim before
release:
citations
spot-
checked,
numbers re-
derived
where
possible,
attributions
tested
against
quote-
attribution
research.

Three writing
rules bind
every
section.

Numbers

carry
citations —
a figure with
no bracketed
source does
not appear.

Claims

carry states
— TRUTH
(verified
against the
cited source),
FALSITY (a
documented
myth or
overclaim),
IGNORANCE
(open, with
an explicit
request for
the data that
would settle
it). **The**

future
section
extrapolates;
it does not
prophecy —
it projects
only
regularities
that have
already
survived
measurement,
and where
published
expert
estimates

disagree, it
reports the
disagreement
as a range
rather than
resolving it
by
preference.

Limitations

— read before citing this paper onward

- **Survivorship bias is structural.**

The history of invention is written from inventions that worked; the denominator of failed attempts is mostly unrecorded, so success-factor claims are weaker than they look.

- **Formal metrics undercount informal invention.**

Patents and R&D budgets measure institutionalized, patentable invention; tacit know-how, process tweaks, and open-source contribution are systematically underweighted.

- **Patent counts confound output with**

• **Source-access bias.** English-language and open-access sources are overrepresented; some primary datasets sit behind paywalls and were verified through their publishers' own summaries.

Sources

Primary and authoritative sources, as verified on 2026-09-03. URLs are given in full so every claim can be re-checked.

^[1] OECD/Eurostat, Oslo Manual 2018 (4th ed.) — definition of innovation; "the requirement for implementation differentiates innovation from other concepts such as invention." https://www.oecd.org/en/publications/oslo-manual-2018_9789264304604-en.html · Oct 2018

^[2] E. M. Rogers, Diffusion of Innovations, Free Press, 1962 — S-curve; adopter categories 2.5% / 13.5% / 34% / 34% / 16%. <https://www.ebsco.com/research-starters/technology/diffusion-innovations> · 1962 · 5th ed. 2003

[3] J. A. Schumpeter, The Theory of Economic Development (1911 German; Harvard UP English ed. 1934) — development as the "carrying out of new combinations." <https://www.janfagerberg.org/a-guide-to-schumpeter/> · 1911/1934 · via Fagerberg's guide

[4] J. A. Schumpeter, Business Cycles (McGraw-Hill, 1939) — invention distinguished from innovation; innovation possible without invention and vice versa (standardly cited to vol. 1, p. 84; paraphrased here). <https://www.cambridge.org/core/journals/journal-of-the-history-of-economic-thought/article/abs/18262F86E5F237B8CB1D88D5C450A72> · 1939

[5] OECD, Frascati Manual 2015 — definition of R&D ("creative and systematic work..."); five criteria: novel, creative, uncertain, systematic, transferable/reproducible. https://www.oecd.org/education/publications/frascati-manual-2015_9789264239012-en.html 1 · Oct 2015 · first edition 1963

[6] S.
Kline &
N.
Rosenberg,
"An
Overview
of
Innovation,"
in The
Positive
Sum
Strategy,
National
Academy
Press,
1986,
pp.
275–
305 —
chain-
linked
model:
"Models
that
depict
innovation
as a
smooth,
well-
behaved
linear
process
badly
misspecify
the
nature
and
direction
of the
causal
factors
at
work."
[https://
nap.nati
onalacad
emies.or
g/read/6
12/chapt
er/18](https://nap.nationalacademies.org/read/612/chapter/18) ·
1986

[11] W. B. Arthur, The Nature of Technology: What It Is and How It Evolves, Free Press, 2009 — combinatorial evolution: "Technologies somehow must come into being as fresh combinations of what already exists." <https://stias.ac.za/2019/06/combinatorial-evolution-a-theory-of-evolution-for-technology/> · 2009 · quote via STIAS summary

[17] S. Kauffman, "Investigations," Santa Fe Institute WP 96-08-072 (1996) and Investigations (Oxford UP, 2000) — origin of "the adjacent possible." https://edge.org/conversation/suart_a_kauffman-the-adjacent-possible · 1996/2000

[18] S. Johnson, Where Good Ideas Come From, Riverhead, Oct 5, 2010 — popularized the adjacent possible for innovation. <https://www.amazon.com/Where-Good-Ideas-Come-Innovation/dp/1594487715> · 2010

[19] Newton
(fluxions,
1665-
66)
and
Leibniz
(published
first,
Acta
Eruditorum
1684)
—
modern
consensus:
independent
invention
of
calculus.
[https://
sites.ma
th.rutge
rs.edu/~
cherlin/
History/
Papers20
02/newto
n.html](https://sites.math.rutgers.edu/~cherlin/History/Papers2002/newton.html) ·
17th c.
·
Rutgers
history
essay

[20] J.
Schmookler,
Invention
and
Economic
Growth,
Harvard
UP,
1966
—
demand-
pull:
capital-
goods
patenting
tracked
investment
(railroads).
Later
re-
estimation
weakened
the
correlations
(Research
Policy).
[https://
www.hup.
harvard.
edu/cata
log.php?
isbn=978
067443328
33](https://www.hup.harvard.edu/catalog.php?isbn=9780674433283) ·
1966 ·
critique:
doi
0048-
7333(90)90022-
X

[22] Royal Society — first meeting Nov 28, 1660 after a Gresham College lecture by Wren; "a Colledge for the promoting of Physico-Mathematicall Experimentall Learning"; charter 1662. <https://royalsociety.org/about-us/who-we-are/history/> · 1660

[23] J. Mokyr, A Culture of Growth: The Origins of the Modern Economy, Princeton UP, 2016 — the Republic of Letters as a competitive transnational market for ideas. <https://press.princeton.edu/books/paperback/9780691180960/a-culture-of-growth/> · 2016

[25] P.
Moser,
"How
Do
Patent
Laws
Influence
Innovation?
Evidence
from
Nineteenth-
Century
World's
Fairs,"
American
Economic
Review
95(4):
1214-
1236
—
~15,000
exhibits;
no-
patent
countries
innovated
as
much,
in
different
sectors:
patents
shaped
direction
more
clearly
than
level.
[https://
www.aeaw
eb.org/a
rticles?
id=10.12
57/00028
28054825
501](https://www.aeaweb.org/articles?id=10.1257/0002828054825501) ·
Sep 2005
·
synthesis:
JEP
27(1),
2013

[36] 3M
Post-it
—
Spencer
Silver's
low-
tack
adhesive
(1968),
Art
Fry's
choir-
hymnal
bookmark
(1974),
national
launch
1980.
[https://
www.inve
nt.org/i
nductee
s/spence
r-silver](https://www.invent.org/inductees/spencer-silver)
 · 1968–
1980 ·
NIHF /
3M

[37] G.
Basalla,
The
Evolution
of
Technology,
Cambridge
UP,
1988
—
continuity,
diversity,
novelty,
selection.
[https://
www.camb
ridge.or
g/core/j
ournals/
british-
journal-
for-the-
history-
of-scienc
e/artic
le/abs/D
E0F9AFBE
91E8F3DA
2DA3C242
44ACF30](https://www.cambridge.org/core/journals/british-journal-for-the-history-of-science/article/abs/DE0F9AFBE91E8F3DA2DA3C24244ACF30)
 · 1988

[38] R. Nelson & S. Winter, An Evolutionary Theory of Economic Change, Belknap/Harvard, 1982

— routines as genes; innovation as search; markets as selection.

<https://www.hup.harvard.edu/books/9780674272286> · 1982

[39] R. C. Allen, "Collective Invention," J. Economic Behavior & Organization 4(1): 1-24

— Cleveland (England) blast furnaces: firms freely revealed designs; industry-wide progress over ~20 years.

<https://robert-c-allen.net/wp-content/uploads/2017/12/collectionlinvent.pdf> · 1983

[45] V. Bush, Science, the Endless Frontier, July 1945 — basic research as "the pacemaker of technological progress"; led to the NSF. (Sources conflict on Jul 5 vs Jul 25 delivery; cited as July 1945.) <https://www.nsf.gov/science-matters/75-years-endless-frontier-vision-future-rooted-past> · Jul 1945

[46] G. S. Altshuller — TRIZ, begun 1946; ~40,000 patent abstracts analyzed by 1969 yielding 40 inventive principles and the contradiction matrix; English: Creativity as an Exact Science (1984). Larger patent counts (200k–2M) are community claims, not independently documented. <https://www.aitr.org/altshuller/116-altshuller> · 1946– · Altshuller Institute

[50] Quote
Investigator
—
earliest
documented
"Genius
is 1
per
cent
inspiration
and 99
per
cent
perspiration":
Idaho
Daily
Statesman,
May 6,
1901
(a
2%/98%
variant
appeared
1898;
Harper's
Monthly
ran
Rosanoff's
reminiscence
in
1932).
[https://
quoteinv
estigato
r.com/20
12/12/1
4/genius
-ratio/
· 1901 ·
attribution
research](https://quoteinvestigator.com/2012/12/14/genius-ratio/)

[52] WIPO,
World
Intellectual
Property
Indicators
2025
—
2024:
3.7M
patent
applications
(+4.9%,
fastest
since
2018;
Asia
70.1%);
grants
~2.1M;
19.7M
patents
in
force;
origins:
China
~1.8M,
US
501,831,
Japan
419,132,
Korea
295,722,
Germany
133,485;
trademarks
11.7M
applications
/
93.2M
active
registrations;
industrial
designs
1.22M
applications.
[https://
www.wip
o.int/we
b-public
ations/w
orld-int
ellectua
l-proper
ty-indic
ators-20
25-highl
ights/e
n/patent
s-highli
ghts.htm](https://www.wipo.int/web-publications/world-intellectual-property-indicators-2025-highlights/en/patents-highlights.html)
1 · Nov
18, 2025

[65] WIPO
— PCT
international
applications
2025:
275,900
(+0.7%);
China
73,718
(26.7%),
US
52,617
(19.1%).
[https://
www.wip
o.int/e
n/web/pc
t-syste
m/highli
ghts/ind](https://www.wipo.int/en/web/pct-system/highlights/index)
[ex](#) · [Mar](#)
2026
release

[66] R. M.
Solow,
"Technical
Change
and
the
Aggregate
Production
Function,"
Rev.
Economics
&
Statistics
39(3):
312–
320 —
US
nonfarm
output
per
man-
hour
doubled
1909–
1949;
87.5%
of the
increase
attributed
to
technical
change.
[http://w
ww.piket
ty.pse.e
ns.fr/fi
les/Solo
w1957.pd](http://www.piketty.pse.ens.fr/files/Solow1957.pdf)
[f](#) · [Aug](#)
1957

[70] J.
Scannell,
A.
Blanckley,
H.
Boldon
& B.
Warrington,
"Diagnosing
the
decline
in
pharmaceutical
R&D
efficiency,"
Nature
Reviews
Drug
Discovery
11:
191-
200 —
Eroom's
Law:
drugs
per
inflation-
adjusted
\$1B
R&D
halved
~every
9
years,
1950-
2010
(~80-
fold
decline).
[https://
www.natu
re.com/a
rticles/
nrd3681](https://www.nature.com/articles/nrd3681)
· 2012

[71] M. Ringel, J. Scannell, M. Baedeker & U. Schulze, "Breaking Eroom's Law," Nature Reviews Drug Discovery — the decline broke around 2010 with a modest uptick; better decision quality and genetically validated targets credited. <https://www.nature.com/articles/d41573-020-00059-3> · Apr 2020

[72] J. DiMasi, H. Grabowski & R. Hansen (Tufts CSDD), J. Health Economics 47: 20–33 — pre-approval cost per approved new drug \$2.558B (2013 USD); \$2.87B including post-approval R&D. <https://www.keionline.org/23054> · 2016

[73] Deloitte, Measuring the Return from Pharmaceutical Innovation (15th-16th eds.) — average cost per asset \$2.23B with 5.9% IRR (2024); \$2.67B with 7.0% IRR (2025), GLP-1s the main driver.
<https://www.deloitte.com/ch/en/Industries/life-sciences-health-care/research/measuring-return-from-pharmaceutical-innovation.html> · 2025/May 2026

[75] European Commission JRC, 2025 EU Industrial R&D Investment Scoreboard — top 2,000 corporate R&D investors spent €1,446B in 2024 (>90% of global business-funded R&D); US firms 47.1% of that spend; Amazon newly ranked #1 (~€65B, Scoreboard estimate). <https://iri.jrc.ec.europa.eu/scoreboard/2025-eu-industrial-rd-investment-scoreboard> · Dec 22, 2025

[78] A. M. Knott, "The Trillion-Dollar R&D Fix," Harvard Business Review, May 2012 — RQ (Research Quotient): a firm's estimated output elasticity of R&D. <https://hbr.org/2012/05/the-trillion-dollar-rd-fix> · May 2012

[82] M. A. Lemley, "Rational Ignorance at the Patent Office," Northwestern Univ. Law Review 95(4) — ~5% of patents are litigated *or* licensed for a royalty (~1.5% litigated); "perhaps 95%" never valuable enough for either (US, ca. 2001). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2614 00 . 2001

[83] P. A. David, "The Dynamo and the Computer," *American Economic Review* 80(2): 355–361 — electric motors <5% of factory drive horsepower in 1899, past ~50% only in the early 1920s; gains required factory reorganization (unit drive). <https://ideas.repec.org/a/aea/aecrev/v80y1990i2p355-61.html> · tml · May 1990

[85] E.
Brynjolfsson,
D.
Rock &
C.
Syverson,
"The
Productivity
J-
Curve,"
AEJ:
Macroeconomics
13(1):
333–
372 —
intangible
investment
makes
measured
TFP
first
understate,
then
overstate
progress;
computer-
era
adjustment
implies
TFP
level
15.9%
higher
by
end-
2017.
[https://
www.aeaw
eb.org/a
rticles?
id=10.12
57%2Fma
c.201803](https://www.aeaweb.org/articles?id=10.1257%2Fmae.201803)
86 · Jan
2021

[86] M.
Marx &
A.
Fuegi,
"Reliance
on
Science,"
Strategic
Management
Journal
41(9)
—
share
of
USPTO
patents
citing
scientific
literature
on the
front
page
rose
6.7%
(1976)
→
25.6%
(2018).
[https://
sms.onli
nelibrar
y.wiley.
com/doi/
full/10.
1002/sm
j.3145](https://sms.onlinelibrary.wiley.com/doi/full/10.1002/smj.3145) ·
2020 ·
relianceonscience.org

[88] A.
Arora,
S.
Belenzon
& A.
Patacconi,
"The
decline
of
science
in
corporate
R&D,"
Strategic
Management
Journal
39(1):
3-32
—
share
of US
firms
publishing
research
fell
17%→6%
(1980-
2006)
while
patenting
rose
15%→25%:
"golden
eggs
without
the
goose."
[https://
onlineli
brary.wi
ley.com/
doi/ful
l/10.100
2/smj.26
93](https://onlinelibrary.wiley.com/doi/full/10.1002/smj.2693) ·
2018

[89] N.
Bloom,
C.
Jones,
J. Van
Reenen
& M.
Webb,
"Are
Ideas
Getting
Harder
to
Find?,"
American
Economic
Review
110(4):
1104–
44 —
aggregate
US
research
productivity
down
×41
since
the
1930s
($\approx -5.1\%/yr$);
halves
every
13
years,
so
research
effort
must
double
every
13
years
to hold
growth
constant;
Moore's-
law
doubling
now
takes
>18×
the
researchers
of the
early
1970s.
[https://
www.aeaw
eb.org/a
rticles?
id=10.12
57/aer.2
0180338](https://www.aeaweb.org/articles?id=10.1257/aer.20180338)
· Apr
2020

[90] B. F. Jones, "The Burden of Knowledge and the 'Death of the Renaissance Man'," Review of Economic Studies 76(1): 283–317 — age at first invention rising ~0.6 years per decade (55,000 inventors); specialization and teamwork rising. <https://academic.oup.com/restud/article-abstract/76/1/283/1577537> · 2009

[93] J. D. Farmer & F. Lafond, "How predictable is technological progress?," Research Policy 45(3): 647–665 — exponential cost-decline forecasting validated across 53 technologies. <https://www.sciencedirect.com/science/article/pii/S0048733315001699> · 2016

[94] R. Way, M. Ives, P. Mealy & J. D. Farmer, "Empirically grounded technology forecasts and the energy transition," Joule — probabilistic Wright's-law forecasts backtested on 50+ technologies; fast transition likely saves trillions. [https://www.cell.com/joule/fulltext/S2542-4351\(22\)00410-X](https://www.cell.com/joule/fulltext/S2542-4351(22)00410-X) · Sep 13, 2022

[97] C. I. Jones, "The End of Economic Growth? Unintended Consequences of a Declining Population," American Economic Review 112(11): 3489–3527 — the "Empty Planet" result: with sub-replacement fertility, living standards stagnate in idea-based growth models. <https://www.aeaweb.org/articles?id=10.1257/aer.20201605> · Nov 2022

^[98] A. N. Whitehead, Science and the Modern World (Lowell Lectures), Macmillan, 1925 — "The greatest invention of the nineteenth century was the invention of the method of invention." <https://www.gutenberg.org/ebooks/68611> · 1925 · Project Gutenberg #68611

^[99] Z. Griliches, "Hybrid Corn: An Exploration in the Economics of Technological Change," *Econometrica* 25(4): 501-522 — "Hybrid corn was the invention of a method of inventing." <https://www.jstor.org/stable/1905380> · Oct 1957

[100] I. Cockburn, R. Henderson & S. Stern, "The Impact of Artificial Intelligence on Innovation," NBER WP 24449; ch. in The Economics of Artificial Intelligence (2019) — deep learning as a general-purpose "method of invention." <https://www.nber.org/papers/w24449> · Mar 2018

[101] CASP14 press release — AlphaFold 2 recognized as a solution to the 50-year protein-structure-prediction challenge. https://predictioncentre.org/casp14/docs/CASP14_press_release.html · Nov 30, 2020

[111] "A generative AI-discovered TNIK inhibitor for idiopathic pulmonary fibrosis: a randomized phase 2a trial" (rentosertib), Nature Medicine — 71 patients; +98.4 mL FVC (60 mg arm) vs −20.3 mL placebo. <https://www.nature.com/articles/s41591-025-03743-2> · Jun 3, 2025

[112] Insilico Medicine — Phase III trial of rentosertib initiated (~320 patients, 47 centers, 52-week FVC endpoint). <https://insilico.com/news/xmj5n4l091-insilico-initiates-phase-iii-clinical-trial> · Jul 7, 2026

[117] Sakana
AI —
first
fully
AI-
generated
paper
to pass
peer
review
at an
ICLR
2025
workshop
(1 of 3
accepted,
score
6.33;
run
with
organizers'
cooperation;
withdrawn
pre-
publication
by
prior
agreement).
Follow-
up "AI
Scientist"
work
published
in
Nature,
Mar
2026
(see
sakana.ai/ai-
scientist-
nature).
[https://
sakana.a
i/ai-sci
entist-f
irst-pub
licatio
n/](https://sakana.ai/ai-scientist-first-publication/) · Mar
2025 /
Mar 2026
